Internship Reports

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1. General Information

Student Name: Moritz Albrecht Matriculation Number: 2134370

Internship Details

Organization: Bundesdruckerei GmbH

Supervisor: Matthias Matuschka

Working Hours: 20h/week
This corresponds to a part-

Remuneration: 1.304,00€/month

Main Tasks: Process
Management/Reporting, Process

Dates:Start: 01.11.2025 End: Language:

Crediting Details

25 ECTS (12 weeks full time)

Moritz Albrecht, 25.10.2026

Student Date

2. Report

2.1 Detailed Description

What did you do? Describe the organization and your tasks (2-3 pages):

2.2 Personal Reflection

What did you learn? Reflect on your learning experience, personal growth, realizations etc. (2-3 pages):

2.3 Weekly Reports

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Appendix A Supporting Python example for Reports 05–06

Analysing Customer Survey Results in *Excel*

Period: 02.03.2026 to 15.03.2026

Task and context

I worked on the analysis of an anonymised customer survey to help assess how customers experienced the company's services and where improvements might be needed. The responses included standardised ratings, product information and written feedback. My task was to organise these different forms of information in *Excel* and prepare an evaluation that supported internal discussion.

Approach and contribution

I first examined the structure and completeness of the dataset, identifying the response categories available for each question and distinguishing unanswered questions from neutral responses. I then used *PivotTables* to count how many respondents selected each response category, from very dissatisfied to very satisfied, and calculated the corresponding percentages. For the written feedback, I developed thematic categories such as communication, price and product quality. I assigned comments to the relevant themes and counted how often each occurred, allowing a comment to cover more than one theme where appropriate. This made the feedback easier to compare while retaining the original wording for interpretation.

Since the responses were anonymized a separate task was to consider whether the available information suggested which customer had provided a response. I used product information and contextual clues in the written comments to infer possible customers. However, these clues generally did not establish their identity reliably, so the exercise largely depended on tentative guesses rather than confirmed assignments.

Outcome and skill development

The evaluation brought together quantitative patterns and qualitative feedback in a form that could support discussion of customer needs and possible improvements. It also made the basis and limitations of the findings more transparent.

This task applied statistical reasoning from my studies to a practical dataset. I developed my ability to interpret response distributions, handle missing information and assess how real world

subtleties, such as product differences, affected numerical comparisons. Analysing written responses also strengthened my ability to develop consistent categories while preserving differences in what respondents meant.

Attempting to identify customers highlighted the distinction between information recorded in a dataset and interpretations based on contextual clues. It showed why a plausible assignment should not be treated as a confirmed finding when the available evidence was limited.

Managing Business Opportunities through *JIRA*

Period: 17.11.2025 to 30.11.2025

Task and context

My daily work at Bundesdruckerei involved screening public procurement notices, identifying potential business opportunities and coordinating their assessment through the company's opportunity pipeline. The objective was to connect public-sector demand with the company's products, expertise and organisational responsibilities. This report focuses on a selected two-week period of these recurring responsibilities.

Approach and contribution

When reviewing a procurement notice, I assessed the requested solution and its relationship to Bundesdruckerei's portfolio. This required interpreting the administrative or technical problem behind the description. For example, a notice seeking secure identification for an online public service raised several distinct questions: did users need to prove their identity when registering, authenticate themselves on later visits, or sign documents electronically? Identity verification using the online ID function pointed towards *AusweisIDent*, offered by our sister cooperation D-Trust and Governikus; a requirement for electronic signatures pointed towards *sign-me*. Clarifying these functions helped me identify the relevant product area

I then assigned relevant opportunities to the appropriate key account management (KAM) team. The subsequent process involved several distinct assessments. KAM evaluated the economic potential of an opportunity; the responsible business unit assessed its fit with the portfolio and available resources; and the technical unit examined the requirements and initiated the development of a suitable solution where appropriate. My role was to support the correct routing of opportunities and maintain an overview of their progression.

I managed and monitored this process through the *JIRA* ticket tool. Tickets served as structured records linking the procurement information to its current status, responsible unit and subsequent evaluation. Monitoring the pipeline required following both individual opportunities and the handovers between teams. Accurate assignment and consistent documentation were important because later decisions depended on information gathered at earlier stages.

Outcome and skill development

My contribution supported a structured connection between market research and the company's commercial and technical evaluation processes. It made opportunities available to the relevant decision-makers and helped maintain traceability over their progress. This work developed my ability to distinguish commercial attractiveness from technical feasibility. It also strengthened my understanding of how a digital workflow represents an organisational process: categories, responsibilities and status transitions must correspond to the decisions people actually make.

Reporting on the Opportunity Pipeline

Period: 12.01.2026 to 25.01.2026

Task and context

Alongside the daily management of opportunities, I carried out the half-yearly analysis of the pipeline. The aim was to assess how successfully identified opportunities progressed through the organisation and to determine where delays occurred. This report examines one instance of that recurring reporting process.

Approach and contribution

I exported the relevant *JIRA* records into *Excel*. I first created an overview of opportunity volumes and outcomes to establish an initial picture of pipeline activity and success rates. I then examined the individual stages: identification of an opportunity, acceptance by the key account manager, assessment by the business and technical units, and preparation of an offer.

This more detailed analysis helped distinguish the volume of incoming opportunities from their progression through the evaluation process. A large number of identified opportunities did not necessarily imply a corresponding increase in offers. Examining the stages separately made it possible to investigate where and why opportunities were rejected, remained pending or advanced to the next step.

My analysis combined quantitative measures with qualitative information. I compared opportunity volumes, progression rates and the time spent at each stage. I also grouped the recorded reasons for rejection into recurring themes, such as portfolio fit, economic attractiveness or resource constraints. Counting these themes helped to show how frequently particular concerns were recorded, while retaining the original comments preserved the context. Together, these analyses identified stages where delays accumulated and helped explain why some opportunities did not progress.

Interpreting the results required care. Time spent in a given stage could include both active assessment and waiting, and cases still under review had to be distinguished from completed ones. In our team discussion, we considered incomplete status updates, handovers between units and competing priorities as possible explanations for delays. We also discussed whether the change of

federal government and uncertainty over public budgets might have influenced procurement timing or opportunity volumes.

Outcome and skill development

I presented the report within our team, where we discussed the findings and possible explanations in detail. My superiors also incorporated the analysis into stakeholder presentations. The results supported process optimisation by making delays and recurring rejection reasons easier to examine.

This task developed my skills in data preparation, aggregation and process analysis. I learned to move from an overall indicator to the underlying stages needed to explain it, and to distinguish an observed pattern from a causal explanation. The work demonstrated how operational data can inform both management communication and practical organisational improvements.

Adding Data Visualisation to Opportunity Reporting

Period: 01.06.2026 to 14.06.2026

Task and context

On my own initiative, I extended the opportunity reporting with visualisations that made developments over time and differences between pipeline stages easier to understand. The reporting already contained numerical information, but separate tables required viewers to reconstruct comparisons themselves. My aim was to make relevant patterns more accessible for analysis and stakeholder presentations.

Approach and contribution

I merged the available historical figures into one dataset, with consistent categories for reporting periods and pipeline stages. Drawing on my Information Visualisation lecture, I began with the questions viewers needed to answer: how had activity changed over time, how did the stages compare, and where did unusually long processing times occur? I treated visualisation as a way of communicating these relationships and guiding interpretation.

For historical developments, I used line charts with half-year reporting periods on the horizontal axis and opportunity counts on the vertical axis. Separate series represented selected pipeline stages. This made it easier to see whether changes in incoming opportunities were accompanied by similar changes in positively assessed cases or offers.

I also considered grouped bar charts for direct comparisons between stages and periods. Lines emphasised temporal development, while bars made individual quantities easier to compare. The choice depended on the question the presentation needed to answer.

The design applied the lecture's distinction between marks and visual channels. Position encoded time and quantity, while consistent colour distinguished the stages. I used clear axis labels, a shared scale and a limited number of series to reduce clutter. Direct labels helped viewers identify the lines without repeatedly consulting a separate legend.

The duration analysis also showed why averages needed context: one *JIRA* ticket that had remained unattended for almost a year strongly affected the reported figures. I therefore

considered box plots as a further extension, since they give a more nuanced picture of outliers and median values.

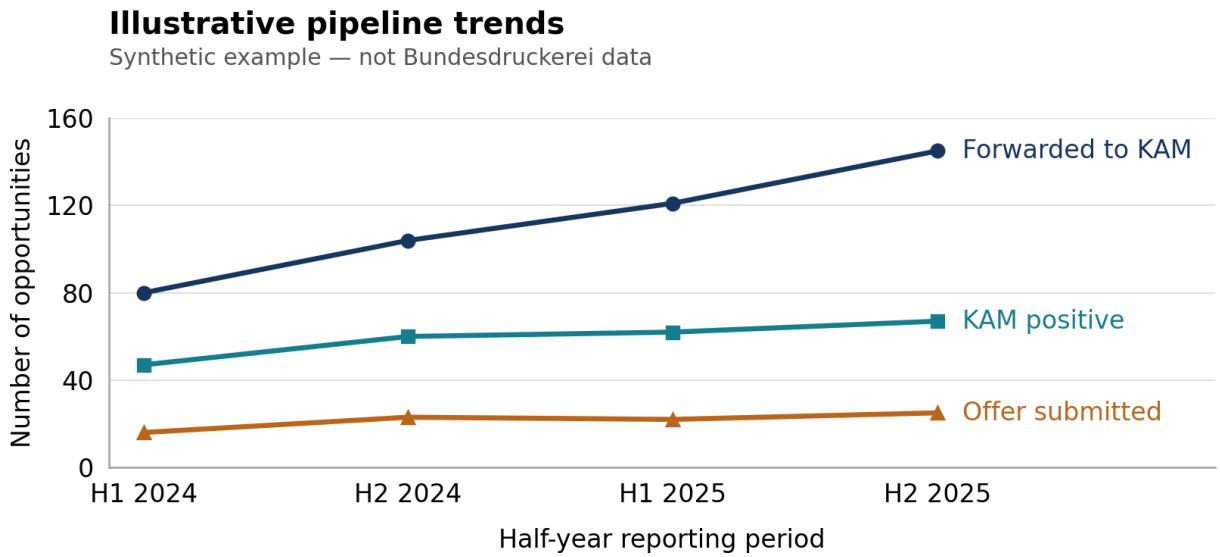


Figure 1. Example of the chart design using synthetic values. Counts refer to separate reporting periods, not a single cohort.

Outcome and skill development

My initiative added a historical and comparative perspective to the reporting and made its findings easier to communicate in presentations. It connected the analysis of the pipeline with the needs of the people interpreting the results.

The task directly applied knowledge from my Information Visualisation course. I developed my ability to select representations for specific analytical questions and to explain how design choices influence which patterns viewers notice.

Designing a Workflow for Automated JIRA Ticket Preparation

Period: 16.03.2026 to 29.03.2026

Task and context

Recording procurement opportunities in JIRA involved transferring information from notices and routing tickets to the responsible key account management (KAM) unit. After colleagues covered my task during an illness, I noticed inconsistent entries. I investigated whether a small automation could make preparation more consistent.

Approach and contribution

I limited the proposal to initial data entry from a standardised procurement PDF (Bekanntmachung). The target fields were client (Auftraggeber), a financial reference for expected revenue (Erwarteter Umsatz), award criteria (Zuschlagskriterien) and duration (Laufzeit). Commercial and technical assessments would remain with the responsible teams.

A notice could contain estimated values, framework ceilings, awarded totals and price weightings. These needed to remain distinct. A colleague would confirm which figure, if any, was suitable for expected revenue. Missing information, multiple lots or conflicting figures would require review.

For routing, I proposed looking up the client in an existing Excel mapping of customers to KAM units. Only an unambiguous match would supply an assignment; unmatched customers or conflicting entries would need clarification.

The prepared values could then be transferred to the JIRA form. A colleague would check them, confirm routing, attach the source document and submit the ticket. The existing KAM, business-unit and technical assessments would follow. I presented this limited workflow and its review points to my superior.

Outcome and skill development

The proposal aimed to reduce repeated copying and searching. It developed my requirements-analysis skills by translating a workplace problem into defined inputs, processing steps and review conditions, and identifying a manageable improvement within a larger process.

Preparing a Python Script to Fill JIRA Ticket Fields

Period: 30.03.2026 to 12.04.2026

Task and context

Following Report 05, I explored how Python could support initial ticket preparation: reading a procurement notice, finding the responsible KAM unit and preparing JIRA fields for review. This remained a small supporting step in the opportunity workflow.

Approach and contribution

I divided the proposed implementation into document extraction, Excel lookup and form transfer. Extraction would use recurring numbered TED sections and their text. It needed to distinguish the buyer from suppliers and retain the context of financial figures, rather than assume that the internal JIRA labels appeared as PDF headings.

The Excel lookup would return a KAM unit only for a clear customer match. Missing fields, repeated sections and conflicting assignments would require clarification. A procurement value could support a draft financial entry, but would not by itself establish our company's expected revenue.

I considered browser automation for transfer, subject to checking the company's actual JIRA field selectors and KAM controls. A colleague would sign in, review and correct the entries, confirm assignment, attach the PDF and submit manually.

Outcome and skill development

The IT department did not approve implementation, so the project stalled before I could test a connection to the live JIRA system. The work therefore remained a proposal and programming example, rather than a deployed company solution.

The task applied functional decomposition, text processing and conditional checks from my studies. It taught me to make format assumptions and human review points explicit, and showed how organisational approval can determine whether a programming proposal progresses.

Appendix A contains a runnable example prepared retrospectively using later public notices and a dummy KAM mapping. It illustrates the proposal separately from the internship outcome. The browser extension remains unverified in the company's JIRA environment.

Automating Newsletter Archiving and Daily Digest Preparation

Period: 13.04.2026 to 26.04.2026

Task and context

Daily *Tagesspiegel Background* (TSB) newsletters provided relevant information for market research at Bundesdruckerei. Saving the editions and preparing them for internal circulation involved repeated manual work. I worked on automating these steps so that colleagues could access the newsletters through a consistent archive and a consolidated daily email.

Approach and contribution

I divided the process into two workflows: archiving each newsletter when it arrived and preparing a daily digest at a scheduled time. A forwarding rule could redistribute individual emails, but it would not provide the PDF archive and consolidated overview needed for subsequent research.

Using *Power Automate*, I configured a workflow that recognised incoming TSB newsletters through their sender and subject. It converted the email's HTML content into a PDF and saved it in a designated SharePoint folder. A naming convention combining publication date and topic, such as 2026-06-15_TSB_Digitalisierung_und_KI.pdf, made the archived editions easier to identify and retrieve.

The conversion required checking whether the resulting PDF preserved the newsletter's content and remained readable. I used the HTML version rather than plain text to retain more of the original structure and formatting, and checked sample PDFs against the source emails.

The second workflow collected the day's archived editions and assembled a consolidated email draft for review. Separating this scheduled step from the arrival-triggered archiving allowed newsletters received at different times to be included in one digest. I checked the draft for completeness before sending it to colleagues.

Outcome and skill development

The workflows reduced the manual effort involved in saving newsletters and preparing their daily circulation. The SharePoint archive provided a consistent location for subsequent research, while the consolidated email made the day's editions available together.

This task developed my ability to translate a recurring activity into explicit triggers, conditions and processing steps. It applied algorithmic thinking from my studies through the sequencing of actions and repeated processing of incoming newsletters.

Evaluating *Assistent.iQ* for Research Support

Period: 15.06.2026 to 28.06.2026

Task and context

I explored how Bundesdruckerei's internal AI assistant, *Assistent.iQ*, could support recurring research tasks. The objective was to improve the consistency of document summaries and preliminary assessments through structured instructions and examples. My contribution concerned developing and evaluating prompts for the assistant, rather than retraining its underlying model.

Approach and contribution

I began by selecting a narrowly defined task: producing a structured briefing from a supplied research document. The expected output included the main development, the organisations involved, relevant dates and potential business implications for Bundesdruckerei.

Initial attempts showed why a general instruction such as "summarise this document" was insufficient. An answer could be fluent while omitting an important qualification or presenting a possible implication as an established fact. I therefore developed a reusable prompt specifying the structure of the answer and requiring a distinction between source-supported information and interpretation.

I added examples demonstrating the desired output and how to handle missing information. This applied the distinction between instruction-only and few-shot prompting from my Intelligent User Interfaces (IUI) lecture on prompting-based interaction. If the document did not establish an implementation date or procurement status, the examples showed an explicit "not stated" response. I also requested references to relevant passages so that important claims could be checked.

To evaluate how well the prompts supported this task, I compared outputs against the original documents using a simple assessment table. The criteria included factual accuracy, coverage of relevant information, unsupported claims and usefulness for the intended reader. I revised the instructions in response to recurring errors and checked them on additional documents that had not guided the initial revisions.

Outcome and skill development

The work resulted in a reusable prompt template and an evaluation approach for AI-assisted research. It developed my ability to turn a broad expectation such as “a useful summary” into observable criteria and to revise instructions through systematic comparison. The task also made the IUI course’s emphasis on user control concrete: useful assistance depended on making outputs easy to verify and correct.

Researching the *Deutschland Stack* and Its Potential Applications

Period: 03.11.2025 to 16.11.2025

Task and context

I researched the *Deutschland Stack* as a proposed shared foundation for digital public services across federal, state and municipal administrations. A central aim was greater digital sovereignty: public institutions should retain control over their data and the systems on which their services depended. During this two-week period in November 2025, I examined the available plans, including the October 2025 federal *Modernisation Agenda*, and assessed where Bundesdruckerei's competencies could contribute.

Approach and contribution

I examined how the initiative connected common standards, reusable components and public-sector applications. Its scope was still contested: in a July 2025 *Tagesspiegel Background* commentary, Thilak Mahendran wrote, "Der Deutschland-Stack ist zur Projektionsfläche geworden." I therefore distinguished political objectives, the architecture governing how components interact, and the actual services proposed for implementation. The October 2025 federal *Modernisation Agenda* provided a more concrete basis for this assessment.

Planned components in October 2025

Selected elements of the federal *Modernisation Agenda*; this is not a final architecture or a list of completed deployments.

Component	Intended role
Digital identity and access	Digital identification, access management and *EUDI Wallet* integration.
Payments	Reusable payment functionality for administrative services.
Once-only data exchange	*NOOTS* to support reuse of administrative evidence and data.
Cloud and service marketplace	*DVC* infrastructure and a marketplace for shared services.
Planning and approvals	An AI-supported platform, with hydrogen infrastructure as a use case.

I related the *Stack* to service reuse (Nachnutzung) and the One-for-All (*Einer-für-Alle*; EfA) model used in implementing the *Online Access Act* (*Onlinezugangsgesetz*; *OZG*), under which one state or a group of states develops and operates a service for use by other administrations. I identified a potential coordination tension: shared *Stack* components could reduce duplication, while existing EfA services already had their own operating arrangements and interfaces.

As a concrete case, I examined the Hamburg-led platform for planning and approval processes for the hydrogen core network. The project had been announced in 2024, and an AI-supported planning and approval platform was included among the *Stack's* proposed components. I considered how document submission and administrative review might connect to shared identity, data-exchange and infrastructure services. These were possible integration paths; the plans did not establish that the platform was already fully integrated into the *Stack* during the research period.

These proposed shared services led to a longer discussion with my superiors about the *Stack's* goal of digital sovereignty. If an authority relied on a common identity or cloud service, could it retain control of its data and change suppliers without rebuilding the entire application? Open interfaces could make components easier to replace, while reusable software could reduce dependence on a single vendor. We also considered the practical limits: an authority still needed the expertise and contractual rights to operate, maintain or replace the technology. This gave the term "sovereignty" a concrete meaning in our assessment.

For Bundesdruckerei, I considered possible contributions in digital identities, secure access and trusted information exchange. I distinguished announced components from available services and concrete procurement opportunities when assessing the maturity of these possibilities.

Outcome and skill development

The research connected the *Stack's* objectives with planned components, a specific application and possible areas of company involvement. It provided a basis for monitoring responsibilities, interfaces and implementation decisions.

This work strengthened my ability to analyse the relationship between infrastructure, shared services and applications. I applied philosophical skills in clarifying digital sovereignty and distinguishing conceptual ambiguity from practical coordination problems. It also showed why technical proposals need to be assessed alongside existing institutional commitments.

Retrospective update

After the period covered by this report, the plans were further refined. On 18 March 2026, the IT Planning Council adopted a platform core covering identity and trust, data exchange, data retrieval, payments and a mailbox, alongside standards for *Stack* solutions.

Researching Federal IT Consolidation

Period: 01.12.2025 to 14.12.2025

Task and context

I researched the German federal IT consolidation programme to assess its implementation status, organisational responsibilities and potential relevance to Bundesdruckerei. The assignment involved understanding both the overall programme and individual digitalisation measures. The objective was to establish where administrative needs and ongoing developments might provide opportunities for the company.

Approach and contribution

I structured the research around the consolidation of IT services, the consolidation of IT operations and the bundling of procurement. This helped distinguish administrative applications, their supporting infrastructure and the arrangements for purchasing them. I also recorded the dates of the underlying documents, because the original programme structure did not necessarily reflect later institutional changes.

I documented individual measures in *Excel*, including their type, supporting sources, legal or administrative basis, responsible institutions and implementation status. Examples included electronic file management, federal payment services, procurement applications and shared infrastructure such as the *Bundescloud*.

An important challenge was establishing which information remained applicable. The creation of the Federal Ministry for Digitalisation and Government Modernisation (BMDS) in the new term meant that older references to ministries or departments needed to be checked against more recent responsibilities. I compared earlier programme documentation with newer information about the BMDS, the Federal Ministry of the Interior (BMI) and ITZBund, distinguishing a historical programme assignment from a current institutional responsibility.

I also distinguished between rollout, ongoing operation and further development. A programme being formally completed did not necessarily mean that implementation needs had disappeared. This distinction informed the assessment of possible Bundesdruckerei involvement: the relevant question was which concrete requirements remained and how they related to the company’s capabilities.

Outcome and skill development

The research resulted in a structured overview of consolidation measures, responsibilities and implementation questions. It provided a basis for identifying areas that warranted further investigation and for directing subsequent research towards the relevant institutions.

The task strengthened my ability to synthesise information from heterogeneous sources and assess its reliability and temporal relevance. It connected technical understanding of IT systems with institutional analysis. I applied conceptual and analytical skills from my studies to distinguishing programme objectives, technical functions and organisational responsibilities, and to explaining how these relationships affected the interpretation of a potential business opportunity.

Comparing Online Identity Verification Services

Period: 15.12.2025 to 04.01.2026

Task and context

I compared services for online identity verification in Germany, focusing on public-service portals. The reference product was *AusweisIDent*, developed and jointly offered by D-Trust, part of the Bundesdruckerei Group, and Governikus. I compared it with *WebID's eID* and *IDnow VideoIdent* from *IDnow GmbH* to examine which customer requirements favoured each approach.

Approach and contribution

I first defined the function being compared: confirming a person's identity remotely using an official identity document. This kept the comparison focused on identification during registration or an application. Routine account login and electronic signing were separate requirements; for example, D-Trust's *sign-me* could support a subsequent signature step, but would not replace the identification service.

I recorded the products in a comparison matrix covering the identification method, documents and equipment required, integration into an existing service and the steps the user had to complete. The main differences were as follows.

Service	Identification method	Practical requirement
*AusweisIDent Online*D-	Reads identity data from the eID chip;	Compatible eID document, PIN and reader
WebID eID	Reads the eID chip through the *My*	Compatible eID document, PIN and NFC
IDnow VideoIdent	A specialist checks the person and	Supported identity document, camera and

AusweisIDent offered an advantage over a video-based approach where users had a suitable eID document and PIN: identity data could be read from the chip without a live call. The service also avoided the need for the portal operator to obtain its own authorisation certificate. However, chip-based identification was not a unique selling point against *WebID's eID*, which used the same underlying online ID function. That comparison required closer attention to the app journey, integration effort, support and commercial terms.

For a portal serving users without an activated eID or available PIN, a video-based alternative could extend access, provided their documents were supported. This showed why the preferred solution depended on the intended users and the procedure's requirements. I could compare

documented functions, but public information did not support a reliable ranking of negotiated prices, completion rates or actual integration costs.

I summarised these trade-offs for internal discussion. The strongest case for *AusweisIDent* was as an eID component within an existing service whose users could use the online ID function. I treated that as a specific use case rather than a claim that it was superior to every competing identification method.

Outcome and skill development

The comparison clarified the relevant market and gave the team specific questions for assessing opportunities: which documents would users have, which identification method was required and how would the service fit into the customer's existing application?

This task combined economic reasoning about competing services with technical analysis of how they worked. I learned to distinguish a product's documented features from a customer advantage, and an advantage shared by several providers from a genuine differentiator.

Researching the *AI Act* and Its Implications for Digital Solutions

Period: 18.05.2026 to 31.05.2026

Task and context

I researched the European *AI Act* to assess its relevance to the development and use of AI-based solutions and to identify potential implications for Bundesdruckerei. The objective was to understand how regulatory requirements could affect public-sector applications and create demand for technical and organisational support.

Approach and contribution

My research centered on the questions that needed to be answered before assessing an AI application's relevance to Bundesdruckerei: What would the system be used for, which regulatory requirements would follow from that use, and who would be responsible for meeting them? I examined these questions using *Assistent.iQ*, Bundesdruckerei's document-based AI assistant for public authorities, as a concrete example. Drawing on the *AI Act* lectures in my Machine Ethics course, I considered how its regulatory assessment depended on the task an authority wanted it to perform. For example, using the assistant to summarise administrative guidance on public benefits would serve a different purpose from asking it to recommend whether a particular applicant should receive a benefit. I treated the latter as a hypothetical extension requiring a separate assessment of the intended use and potentially relevant high-risk requirements. This helped me examine the boundary between the product's document-assistance functions and possible customer requests for support with individual decisions.

I then considered how responsibilities would be divided between Bundesdruckerei as the product provider and a public-sector customer using the assistant. Continuing the public-benefits example, I considered a hypothetical administrative office using *Assistent.iQ* to summarise guidance when preparing replies to citizens. Bundesdruckerei would need to explain the product's intended functions and limitations, while the office would need to establish how staff should use and check its outputs. For example, the product's source references could help a caseworker compare an AI-generated summary with the underlying guidance before including it in a reply. The office would still need to decide who performed this check and who remained responsible for the final

response. This connected my research to a hypothetical customer need: introducing the product required both verifiable outputs and clear responsibilities within the customer's workflow.

Outcome and skill development

The research gave me a more precise basis for discussing AI applications and potential customer needs with the team. It connected regulatory requirements to questions about the intended workflow and the company's capabilities.

The task applied the contents of the Machine Ethics lectures by examining how the discussed concepts, like transparency, oversight and accountability, were required in practice. The main skill I developed was translating abstract regulatory and ethical concepts into concrete questions about product functionality and administrative workflows. In particular, I learned that providing source references supports verification, while effective oversight also requires clear responsibilities for checking and using the output.

Preparing procurement fields for human review

A.1 Scope and use

This example was prepared retrospectively to illustrate the limited automation proposed in Reports 05–06. The five sample notices were published in October 2026, after the internship period. The implementation and checks described here are not presented as activities completed during the March–April reporting periods or as an approved company deployment.

The script reads text-based German TED PDFs, extracts the buyer, duration and award criteria, and looks up a KAM unit in Excel. It prints the five target fields and saves a JSON draft containing source evidence and review issues. Missing or ambiguous values remain unresolved; they do not erase the fields that were successfully extracted. Scanned PDFs and other layouts need additional processing.

The optional --financial-basis argument selects estimated, awarded or ceiling figures. A draft is populated only when the selected category contains one distinct numeric amount and currency, with no multiple lot/part identifiers. All candidate figures remain in the evidence. The selected figure must still be confirmed or replaced before it is used as expected revenue. Without this option, the revenue field stays empty for manual selection.

The active Excel worksheet needs the columns Auftraggeber and KAM. Matching ignores case and extra whitespace; no fuzzy assignment is made. The supplied dummy workbook uses KAM BD, KAM SI, KAM DG and KAM FFO. These are fictional unit assignments, not people or Jira account identifiers.

Install Python 3.10+ and the two required libraries, then run from the folder containing procurement.py. The continuation character below is for a macOS/Linux terminal; on Windows, enter the prepare command on one line.

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python procurement.py prepare "2026-OJS202-00719016-de.pdf" \
  --mapping examples/kam_mapping_dummy.xlsx \
  --financial-basis awarded --output output/lka-demo
```

Use a new output directory for each run: existing reports are not overwritten. A successful exit means that reports were written, not that their contents were approved. The companion ZIP includes the source, requirements, dummy workbook, tests and configuration example.

A.2 Worked example and validation

For the LKA award notice 719016-2026, the tested example produced the following draft. Award notices illustrate extraction; they do not establish that an already-awarded contract is a new sales opportunity.

```
Auftraggeber: Landeskriminalamt Nordrhein-Westfalen
Erwarteter Umsatz: 277 850,75 EUR
Zuschlagskriterien: Preis; Gewichtung 100 Punkte
Laufzeit: Andere Laufzeit: Unbegrenzt
KAM: KAM SI
REVIEW: Draft reference value only; confirm expected revenue
        before Jira transfer.
```

The award-criteria line above is abbreviated for readability; the JSON report and terminal output retain the full extracted criteria block. The selected amount comes from “Wert aller in dieser Bekanntmachung vergebenen Verträge” in section 6. It is a draft reference, not an automatically calculated revenue forecast.

Notice / buyer	Dummy unit	Observed result
712954 / Procurement, Claim and Mobility	KAM FFO	No financial figure or award criteria; duration explicitly unknown.
719016 / LKA Nordrhein-Westfalen	KAM SI	Awarded reference 277 850,75 EUR; price criterion and unlimited duration.
727655 / Landeshauptstadt München	KAM DG	Conflicting estimates retained for review; duration 48 months; criteria refer to procurement documents.
733448 / gematik GmbH	KAM BD	No structured financial figure or award criteria; base duration 24 months plus extension evidence.
737139 / Stadt Bruchsal	KAM DG	Awarded reference 66 125,84 EUR; price criterion; start and end dates extracted.

All five supplied PDFs were processed with the dummy mapping. Seven automated tests passed, covering buyer/supplier scope, financial categories versus price weightings, missing and conflicting values, multiple lots, and unique, missing or conflicting KAM matches. These checks establish the limited offline demonstration; they do not validate real company assignments or a live Jira connection.

A.3 Complete Python listing — procurement.py

The following is the complete runnable script included in the companion package. The default prepare command is offline. The fill-jira command is an explicit, optional extension and contains no automatic submission action.

```
"""Prepare reviewable Jira field drafts from German TED PDF notices."""

from __future__ import annotations
import argparse
import hashlib
import json
import re
import unicodedata
from decimal import Decimal
from pathlib import Path
from pypdf import PdfReader
from openpyxl import load_workbook

HEADING = re.compile(r"^(\d+(?:\.\d+)*\.)\s+(.+)$")
MONEY = re.compile(
    r"^(Geschätzter Wert ohne MwSt\.|Höchstwert der Rahmenvereinbarung|Wert aller in dieser
Bekanntmachung vergebenen Verträge|Wert des Angebots):\s*(.+)$"
)
FINANCIAL_KINDS = {
    "estimated": "Geschätzter Wert ohne MwSt.",
    "awarded": "Wert aller in dieser Bekanntmachung vergebenen Verträge",
    "ceiling": "Höchstwert der Rahmenvereinbarung",
}
AMOUNT = re.compile(r"^((?:\d{1,3}(?:[ .]\d{3})+|\d+)(?:,\d{2})?)\s+([A-Z]{3})$")


def normalize(value):
    return " ".join(unicodedata.normalize("NFKC", str(value)).casefold().split())


def read_sections(path):
    sections, current = [], None
    reader = PdfReader(path)
    for page_no, page in enumerate(reader.pages, 1):
        for raw in (page.extract_text(extraction_mode="layout") or "").splitlines():
            line = raw.strip()
            if (
                not line
                or re.match(r"^\d+-\d{4}\s+Page \d+/\d+$", line)
                or line == "See the notice on TED website"
            ):
                continue
            match = HEADING.match(line)
            if match:
                current = {
                    "section": match[1],
                    "heading": match[2],
                    "page": page_no,
                    "lines": [],
                }
                sections.append(current)
            elif current:
                current["lines"].append(line)
    return sections


def evidence(section, value):
    return {
        "value": value,
        "section": section["section"],
        "heading": section["heading"],
        "page": section["page"],
    }


def lookup_kam(path, client):
    workbook = load_workbook(path, read_only=True, data_only=True)
```

```
    try:
        rows = iter(workbook.active.values)
        headers = [normalize(x or "") for x in next(rows, ())]
        if headers.count("auftraggeber") != 1 or headers.count("kam") != 1:
            raise ValueError(
                "Excel needs unique Auftraggeber and KAM column headers in row 1."
            )
        ci, ki = headers.index("auftraggeber"), headers.index("kam")
        matches = [
            row[ki]
            for row in rows
            if row[ci] is not None and normalize(row[ci]) == normalize(client)
        ]
        names = {str(x).strip() for x in matches if x is not None and str(x).strip()}
        if not matches:
            return None, "No exact customer match in Excel."
        if any(x is None or not str(x).strip() for x in matches) or len(names) != 1:
            return None, "Blank or conflicting KAM assignments in Excel."
        return names.pop(), None
    finally:
        workbook.close()


def select_financial_reference(amounts, sections, basis):
    """Use only the explicitly requested category; never infer sales revenue."""
    selection = {"basis": basis, "value": None, "sources": []}
    if basis is None:
        return selection, None
    if basis not in FINANCIAL_KINDS:
        raise ValueError("Unknown financial basis.")
    lots = {
        m[1]
        for s in sections
        if (m := re.match(r"^(?:Los|Teil):\s*(\S+)", s["heading"]))
    }
    if len(lots) > 1:
        return (
            selection,
            "Multiple lots or parts; select the financial reference manually.",
        )
    values = {}
    for candidate in amounts:
        if candidate["kind"] != FINANCIAL_KINDS[basis]:
            continue
        text = unicodedata.normalize("NFKC", candidate["value"])
        match = AMOUNT.fullmatch(text)
        if match:
            number = Decimal(
                match[1].replace(" ", "").replace(".", "").replace(",", ".")
            )
            values.setdefault((number, match[2]), []).append(candidate)
    if not values:
        return (
            selection,
            f"No published numeric {basis} value with currency; check manually.",
        )
    if len(values) != 1:
        return (
            selection,
            f"Conflicting {basis} values; select the financial reference manually.",
        )
    sources = next(iter(values.values()))
    selection.update(value=sources[0]["value"], sources=sources)
    return (
        selection,
        "Draft reference value only; confirm expected revenue before Jira transfer.",
    )


def extract(path, mapping=None, financial_basis=None):
    sections = read_sections(path)
    issues, clients, amounts, durations, criteria, extensions = [], [], [], [], [], []
    for section in sections:
        lines = section["lines"]
        if section["section"] == "1.1." and section["heading"] == "Beschaffer":
            for i, line in enumerate(lines):
                if line.startswith("Offizielle Bezeichnung:"):
                    parts = [line.split(":", 1)[1].strip()]
```

```
                    for following in lines[i + 1 :]:
                        if ":" in following:
                            break
                        parts.append(following)
                    clients.append(evidence(section, " ".join(parts)))
        for line in lines:
            match = MONEY.match(line)
            if match:
                amounts.append({**evidence(section, match[2]), "kind": match[1]})
        if section["section"].startswith(("3.", "5.")) and section["heading"] in (
            "Dauer",
            "Geschätzte Dauer",
            "Laufzeit",
        ):
            durations.append(evidence(section, "\n".join(lines)))
        if section["heading"] == "Verlängerung":
            extensions.append(evidence(section, "\n".join(lines)))
        if section["heading"] == "Zuschlagskriterien":
            criteria.append(evidence(section, "\n".join(lines)))
    client = clients[0]["value"] if len(clients) == 1 else None
    if not client:
        issues.append("Missing or repeated buyer section; check Auftraggeber.")
    # A notice value is evidence, never an automatic sales revenue forecast.
    for key, values in [
        ("financial figure", amounts),
        ("duration", durations),
        ("award criteria", criteria),
    ]:
        if not values or any(not x["value"] for x in values):
            issues.append(f"Missing {key}.")
        elif key != "financial figure" and len(values) != 1:
            issues.append(f"Multiple {key} sections; select the intended lot manually.")
    if extensions:
        issues.append(
            "Contract extension options exist; review extension evidence separately from base duration."
        )
    selection, financial_issue = select_financial_reference(
        amounts, sections, financial_basis
    )
    if financial_issue:
        issues.append(financial_issue)
    elif amounts:
        issues.append(
            "Confirm financial basis manually: notice estimates, ceilings and awarded values are not
expected revenue."
        )
    if any("Unbekannt" in x["value"] for x in durations):
        issues.append("Contract duration is explicitly unknown.")
    if any("Vergabeunterlagen" in x["value"] for x in criteria):
        issues.append("Award criteria refer to procurement documents; consult them.")
    kam = None
    if mapping and client:
        kam, problem = lookup_kam(mapping, client)
        if problem:
            issues.append(problem)
    elif not mapping:
        issues.append("No Excel KAM mapping supplied.")
    if not sections:
        issues.append(
            "No TED sections found; scanned PDFs require OCR or a different parser."
        )
    return {
        "source": str(path.resolve()),
        "sha256": hashlib.sha256(path.read_bytes()).hexdigest(),
        "status": "manual_review_required",
        "fields": {
            "Auftraggeber": client,
            "Erwarteter Umsatz": selection["value"],
            "Zuschlagskriterien": criteria[0]["value"] if len(criteria) == 1 else None,
            "Laufzeit": durations[0]["value"] if len(durations) == 1 else None,
            "KAM": kam,
        },
        "evidence": {
            "clients": clients,
            "financial_candidates": amounts,
            "durations": durations,
            "extensions": extensions,
            "award_criteria": criteria,
```

```
        },
        "financial_selection": selection,
        "review_issues": issues,
        "reviewed": False,
    }


def fill_jira(report_path, config_path):
    """Explicit opt-in; only text inputs, never submit or select an assignee."""
    from playwright.sync_api import sync_playwright

    report = json.loads(report_path.read_text(encoding="utf-8"))
    config = json.loads(config_path.read_text(encoding="utf-8"))
    fields = report["fields"]
    if (
        report.get("reviewed") is not True
        or report.get("review_issues")
        or any(
            not str(fields.get(k) or "").strip()
            for k in (
                "Auftraggeber",
                "Erwarteter Umsatz",
                "Zuschlagskriterien",
                "Laufzeit",
                "KAM",
            )
        )
    ):
        raise ValueError(
            "Review all fields, resolve review_issues and set reviewed=true before filling Jira."
        )
    from urllib.parse import urlsplit

    if urlsplit(config["url"]).scheme != "https":
        raise ValueError("Jira URL must use HTTPS.")
    selectors = config["selectors"]
    if set(selectors) != set(fields):
        raise ValueError("Configure exactly one text-field selector per report field.")
    with sync_playwright() as p:
        browser = p.chromium.launch(headless=False)
        try:
            page = browser.new_page()
            page.goto(config["url"])
            input(
                "Sign in and open the create form, then press Enter here to fill it: "
            )
            locators = {
                key: page.locator(selector) for key, selector in selectors.items()
            }
            for locator in locators.values():
                if locator.count() != 1 or not locator.is_editable():
                    raise ValueError(
                        "Every selector must identify one editable text field."
                    )
            for key, locator in locators.items():
                locator.fill(str(fields[key]))
            input(
                "Review entries, confirm KAM, attach PDF and submit manually if appropriate. Enter closes
browser: "
            )
        finally:
            browser.close()


def main():
    parser = argparse.ArgumentParser(description=__doc__)
    sub = parser.add_subparsers(dest="command", required=True)
    prepare = sub.add_parser("prepare")
    prepare.add_argument("pdfs", nargs="+", type=Path)
    prepare.add_argument("--mapping", type=Path)
    prepare.add_argument("--output", type=Path, default=Path("output"))
    prepare.add_argument(
        "--financial-basis",
        choices=tuple(FINANCIAL_KINDS),
        help="Explicit draft reference category; always requires revenue confirmation.",
    )
    fill = sub.add_parser("fill-jira")
    fill.add_argument("report", type=Path)
```

```
    fill.add_argument("--config", required=True, type=Path)
    args = parser.parse_args()
    if args.command == "fill-jira":
        fill_jira(args.report, args.config)
        return 0
    args.output.mkdir(parents=True, exist_ok=True)
    failed = False
    for path in args.pdfs:
        try:
            report = extract(path, args.mapping, args.financial_basis)
            target = args.output / f'{path.stem}-{report["sha256"][:12]}.json'
            # Never overwrite a colleague's reviewed report.
            with target.open("x", encoding="utf-8") as stream:
                json.dump(report, stream, ensure_ascii=False, indent=2)
                stream.write("\n")
            print(
                f'{path.name}: {target} ({len(report["review_issues"])} review issues)'
            )
            for field, value in report["fields"].items():
                print(
                    f'  {field}: {value if value is not None else "[not available / needs review]"}'
                )
            for issue in report["review_issues"]:
                print(f"  REVIEW: {issue}")
        except Exception as error:
            print(f"{path}: ERROR: {error}")
            failed = True
    return 1 if failed else 0


if __name__ == "__main__":
    raise SystemExit(main())
```

A.4 Optional Jira transfer and companion files

Offline processing requires no Jira credentials. Browser filling additionally requires Playwright and Chromium, a visible desktop browser, the actual HTTPS Jira address, and selectors checked against the target form. This extension has not been validated in the company's Jira system and would require organisational approval before use.

```
python -m pip install playwright==1.56.0
python -m playwright install chromium
```

Copy jira.example.json to jira.local.json and replace every placeholder. Each selector must identify exactly one editable text field. Jira autocomplete, group and assignee controls may need a project-specific adapter; entering a KAM unit label does not select an account or confirm routing.

```
{
  "url": "https://your-company.atlassian.net/",
  "selectors": {
    "Auftraggeber": "REPLACE_WITH_CUSTOMER_TEXT_SELECTOR",
    "Erwarteter Umsatz": "REPLACE_WITH_REVENUE_TEXT_SELECTOR",
    "Zuschlagskriterien": "REPLACE_WITH_CRITERIA_TEXT_SELECTOR",
    "Laufzeit": "REPLACE_WITH_DURATION_TEXT_SELECTOR",
    "KAM": "REPLACE_WITH_KAM_TEXT_SELECTOR"
  }
}
```

After comparing the JSON draft with the PDF and the authorised customer mapping, correct the fields, resolve the review_issues entries and set reviewed to true. Only then can the optional command fill the configured fields. The colleague signs in normally, reviews the result, confirms the KAM control, attaches the PDF and submits manually.

```
python procurement.py fill-jira output/lka-demo/REPORT.json \
  --config jira.local.json
```

Replace REPORT.json with the actual generated filename. The script opens no browser during prepare, and stores no browser login session. If a field remains unavailable, keep the draft for manual clarification rather than inventing a value to bypass the review check.

Companion files

procurement.py — complete source listed in A.3.

requirements.txt — pinned dependencies: pypdf 6.1.3, openpyxl 3.1.5 and et_xmlfile 2.0.0.

examples/kam_mapping_dummy.xlsx — fictional client-to-unit mapping for the five sample notices.

jira.example.json — configuration placeholders for the unverified browser extension.

tests/test_procurement.py — seven automated tests; run python -m unittest discover -s tests -v.

README.md — installation, processing and review instructions.